

SAFELINE

Metal Detection

Mettler-Toledo Product Inspection
1571 Northpointe Parkway
Lutz, FL 33558

Main: (813) 889-9500
Fax: (813) 881-0840
24/7 Support: (800) 447-4439

www.mt.com/pi



Color Inside

Customer		Documentation			
Name:	Wayne Farms LLC	Designation:	EPL010605	Rev:	A
Country:		EA Number:	30916892		
System		Drawn:	E. Ymeraj		
Name:	M34R V5 Metal Detector SL1500	Drawn Date:	9/13/2023		
Assembly		Checked:	D. Doyle		
SAP Material #:	30916924	Checked Date:	9/13/2023		
Description:	Enclosure, EAC,Main SL1500, EL_ENC_B	SO#			
Complementary Assemblies		SO#_Line Item(s):			
SAP Material #:	Description:	145785007 _10			
30916934	Kit, EAC,Mechanical Misc SL1500	-			
-	-	-			
-	-	-			
-	-	-			

Form PIUS.SL.TITLE_PAG.F261R08

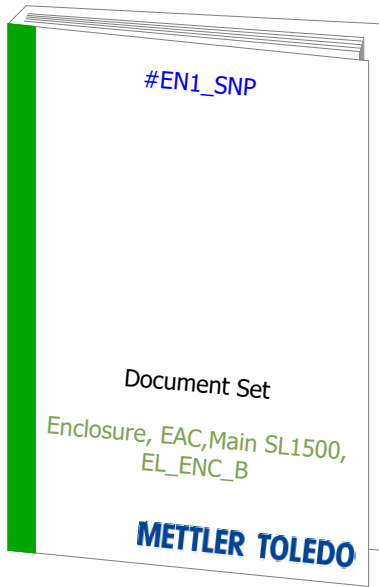
Title page

Cover Sheet

Electrical Schematic

Form PIUS.SL.TAB_CON.F06.R04

Page designation					Page description	Date	Edited by
DCC							
Function							
Mounting location							
Number							



&EFS1

Electrical Schematic

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This safety circuit diagram is an outcome of a comprehensive risk assesment and qualification of the circuit components and architecture. This safety circuit should not be modified in any manner at any time without the express written consent of Mettler-Toledo. Modification of the circuit structure or components may cause damage to property, injury to persons, or death.

Project Name EPL010605		Project Number EPL010605		Change		9/26/2023		Assembly ID 30916892A		ECN No.	
Project Description M34R V5 Metal Detector SL1500				Drawn E. Ymeraj		9/13/2023		Function		Scale 1 : 1	
Variant Description Enclosure, EAC,Main SL1500, EL_ENC_B				Checked D. Doyle		9/13/2023		Mounting Location		Project Status	
Page Description Table of Contents				Document Type Electrical Schematic		DCC &EFS1		Project Rev A		Format ANSI A	
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General Information

COLOR ABBREVIATION	
BK	BLACK
BN	BROWN
RD	RED
OG	ORANGE
YE	YELLOW
GN	GREEN
BU	BLUE
VT	VIOLET
GY	GREY
WH	WHITE
PK	PINK
GD	GOLD
TQ	TURQUOISE
SR	SILVER
GNYE	GREEN/YELLOW
SH	SHIELD
WHBU	WHITE/BLUE
WHBK	WHITE/BLACK
WHBN	WHITE/BROWN

WIRE GAUGE	
mm ²	AWG
0.14	26
0.25	24
0.34	22
0.50	20
1.0	18
1.5	16
2.5	14
4	12
6	10
16	8

CIRCUIT IDENTIFICATION (EXCEPT MULTI-CORE CABLES)	
DESCRIPTION	COLOR
AC power circuits Grounded AC power circuits (Neutral) Protective Earth Conductors <50VDC power & control Grounded <50VDC power & control External ungrounded circuits External grounded circuits	BLACK WHITE GREEN/YELLOW BLUE WHITE/BLUE ORANGE WHITE/ORANGE

DEVICE DESIGNATION	
-B	Sensor e.g. Reject Verification, Bin Full
-BGC	Interlock Switch
-BPA	Air Pressure Switch
-BS	Encoder, Tachometer
-F	Fuse, Miniature Circuit Breaker
-G	Energy source e.g. X-Ray Generator
-I/P	Programmable Input located in MD Power Supply
-K	Relay (except for Power Contactor Relay)
-KF	PC, PLC, HiQiQ Detector
-KFD	Safety Relay
-KG	Relay Timer Multifunction
-KH	Pneumatics Valve e.g. Reject Valve, Air Dump
-KK	Solenoid Operated Lock i.e. Bin Locking Solenoid
-KM	Relay for power isolation
-M	Motor, Fan
-MDPS	Metal Detector Power Supply
-O/P	Programmable Input located in MD Power Supply
-P	Lamp, Signaling Device, Beacon
-PG	Low Air Indicator
-PH	HMI, Text display
-PJ	Horn, Buzzer
-PR	Safety Circuit Warning Indicator
-Q	Main Switch, Isolator/Power Disconnect
-R	Resistor Network PCB, Potentiometer
-RFA	EMC Filter
-RFB	EMC Core (Ferrite)
-RL	Relay located in MD Power Supply
-S	Switch (except air), Pushbutton, Safety Interlock
-SGC	Switch Emergency Stop
-T	Mains filter, Surge Suppressor, Power Supply
-TA	Motor Speed Controller
-U	Mounting panel
-W	Multi-core Cable
-X	Terminal, Connector, Plug & Socket
-XE	Earth point (bonding)
-XG	IO Distribution

TERMINAL TIGHTENING TORQUE			
PART#/SAP#	DESCRIPTION	WIRE GAUGE AWG (mm ²)	TORQUE lb•in (N•m)
28217660	Lug, Grounding/Aluminum	#14-10 (0.82 - 5.3) #8 (8.4)	20 (2.3) 25 (2.8)
28218378	Kit, Ground Bar/5 Terminal/2.54"L	#14-10 (0.82 - 5.3) #8 (8.4)	20 (2.3) 25 (2.8)
28218263	Block, Terminal/PA/Orange	#26-10 (up to 4)	9.0 (1.0)
28120751	Block, Terminal/2-Tier/25A/600V	#26-12 (up to 2.5)	4.5...7.1 (0.5...0.8)

SYMBOL	DESCRIPTION	SYMBOL	DESCRIPTION
	EXTERNAL POWER SOURCE		KEY SWITCH (NORMALLY OPEN)
	PNEUMATICS VALVE		PUSH BUTTON SWITCH (NORMALLY OPEN)
	RELAY COIL		1) SWITCH CONTACT 2) POWER CONTACTOR (NORMALLY OPEN)
	MOTOR (3 PHASE)		PUSH BUTTON SWITCH (NORMALLY CLOSED)
	PROXIMITY SWITCH		1) SWITCH CONTACT 2) POWER CONTACTOR (NORMALLY CLOSED)
	PRESSURE SWITCH (AIR DUMP VALVE)		EMERGENCY STOP PUSH BUTTON (LATCHED)
	PHOTOELECTRONIC SENSOR (RETRO-REFLECTIVE)		MINIATURE CIRCUIT BREAKER
	POWER SUPPLY UNIT		MECHANICAL LIMIT SWITCH
	FILTER		PRESSURE SWITCH
	ROTARY DISCONNECT SWITCH		SIGNAL LAMP
	DEVICE/ ELECTRICAL ASSEMBLY		FUSE
	MULTI-CORE CABLE N= NO. OF CONDUCTORS S= SIZE OF CONDUCTORS		PLUG / SOCKET
	SHIELD/CABLE SCREEN		TERMINAL
	VOLTAGE INDICATOR		

Nameplate

SO#: 145785007 _ 10

Model Type: SL1500

Date of Manufacture:

Voltage: 120 VAC 1Ø

FLA: 15 A

Frequency: 50/60 Hz **Largest Motor or Load (A):** 3.1 A

SCCR: 5 kA rms @ 120 VAC max

Air Supply: -

Electrical Dwg: EPL010605

Rev: A

Enclosure Rating: Type 4X

IP Rating: IP69

Label: No

Safety Circuit



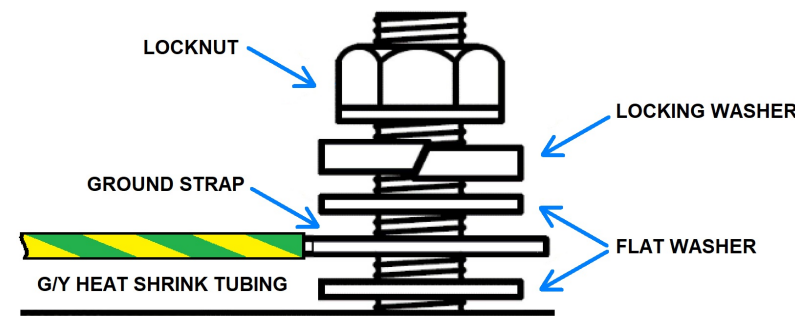
WARNING

The safety circuit diagram is an outcome of a comprehensive risk assessment and qualification of the circuit components and architecture. The safety circuit should not be modified in any manner at any time without the express written consent of Mettler-Toledo. Modification of the circuit structure or components may cause damage to property, injury to persons, or death.

Notes

1. For panels requiring UL 508A, clearly label main disconnect and all circuit breakers and fuses with their values.

Grounding



General Info



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Project Name EPL010605	Project Number EPL010605	Change	6/23/2023	Assembly ID 30916892A	ECN No.
Project Description M34R V5 Metal Detector SL1500	Drawn E. Ymeraj	9/13/2023	Function	Scale 1 : 1	
Variant Description Enclosure, EAC,Main SL1500, EL_ENC_B	Checked D. Doyle	9/13/2023	Mounting Location	Project Status	Format ANSI A
Page Description General Info	Document Type Electrical Schematic	DCC &EFS1	Project Rev A	Page 3 / 16	

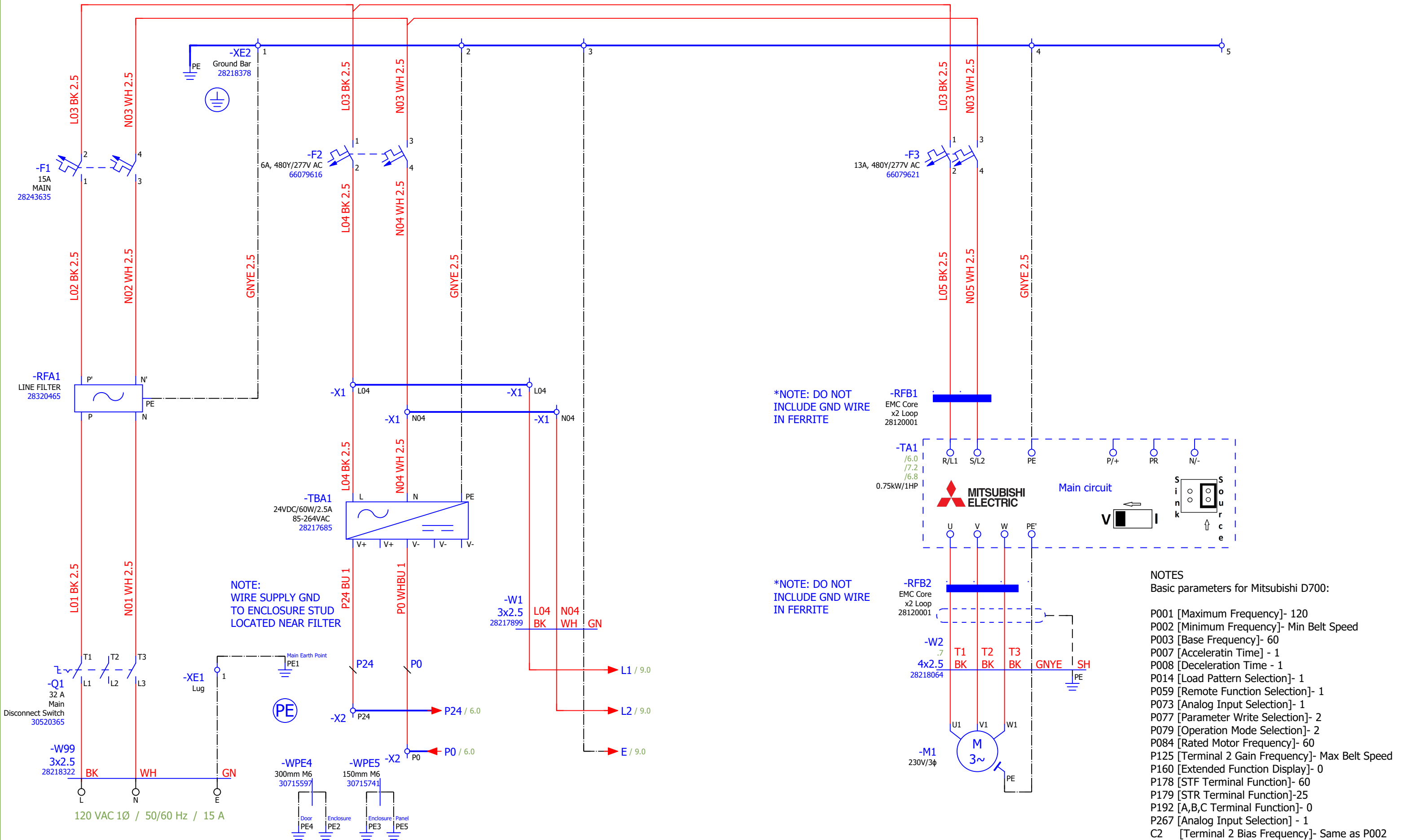
Parts List

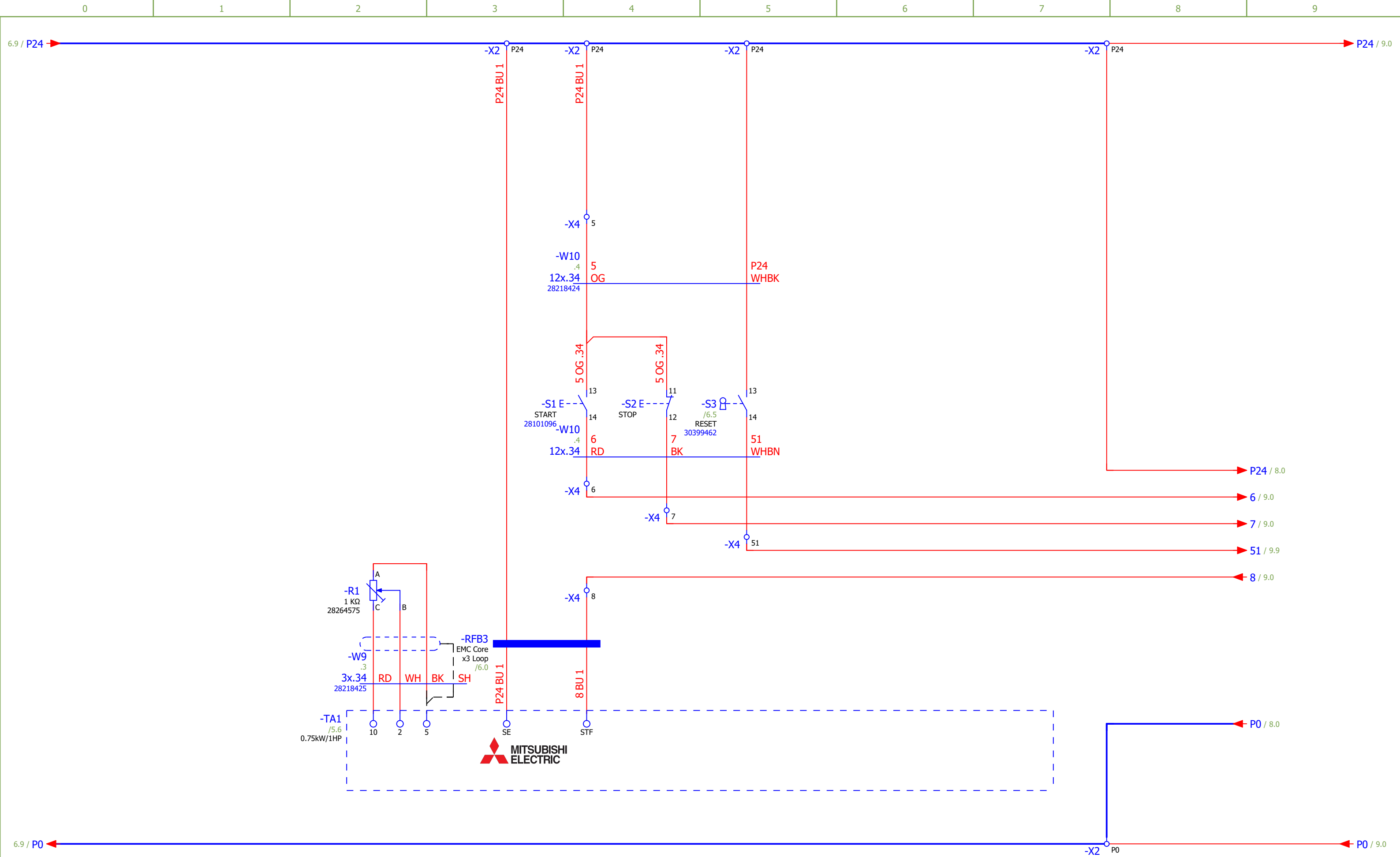
Assembly	Device tag	Qty	UoM	ERP Number	Descripttion	Page No.
Enclosure, EAC,Main SL1500, EL_ENC_B		1	each	28218362	Plug,Vent/M12/0.5"L/Black/Nylon/UL	5
Enclosure, EAC,Main SL1500, EL_ENC_B		1	each	28218361	Fitting,Conduit/Locknut/M12/Vent/Nylon	5
Enclosure, EAC,Main SL1500, EL_ENC_B		1	each	28216936	Assy, Enclosure/All-In-One	5
Enclosure, EAC,Main SL1500, EL_ENC_B	F1	1	each	28243635	Circuit Breaker,15A,C-curve,2Pole,UL489	5
Enclosure, EAC,Main SL1500, EL_ENC_B	F2	1	each	66079616	Circuit Breaker,6A,C-curve,2 Pole,UL489	5
Enclosure, EAC,Main SL1500, EL_ENC_B	F3	1	each	66079621	Circuit Breaker,13A,C-curve,2Pole,UL489	5
Enclosure, EAC,Main SL1500, EL_ENC_B	K1	1	each	66048210	Relay,SPDT/24V/Phoenix	8
Enclosure, EAC,Main SL1500, EL_ENC_B	KFD1	1	each	30332717	Safety Controller,4 input 4 output	6
Enclosure, EAC,Main SL1500, EL_ENC_B	Q1	1	each	30520365	Isolator Base Mounted,32A,3 pole	5
Enclosure, EAC,Main SL1500, EL_ENC_B	Q1	1	each	28217661	Kit,Shaft/Bezel Adapter/Metal/041-6063	5
Enclosure, EAC,Main SL1500, EL_ENC_B	Q1	1	each	28217684	Handle,Disconnect/Oper/Lockable/Disc/AB	5
Enclosure, EAC,Main SL1500, EL_ENC_B	Q1	1	each	30877306	NFPA, Internal Handle (Mod)(All in One)	5
Enclosure, EAC,Main SL1500, EL_ENC_B	Q1	1	each	30528328	Gasket, IP69/for AB Disconnect 28242960	5
Enclosure, EAC,Main SL1500, EL_ENC_B	R1	1	each	28264575	Potentiometer,1K/IP69/Schneider	7
Enclosure, EAC,Main SL1500, EL_ENC_B	RFA1	1	each	28320465	Power Supply,Filter,2ST,16A,Schaffner	5
Enclosure, EAC,Main SL1500, EL_ENC_B	RFA1	1	each	28282215	Plate, Mount/AIO/Filter	5
Enclosure, EAC,Main SL1500, EL_ENC_B	RFB1 ...RFB3	3	each	28120001	Ferrite,380 Ohm,19Dia	5;6
Enclosure, EAC,Main SL1500, EL_ENC_B	S1	1	each	28101096	Pushbutton,Multi-headed ZB4 BA7121	7
Enclosure, EAC,Main SL1500, EL_ENC_B	S1	1	each	28218295	Pushbutton,Boot/Rubber/22mm	7
Enclosure, EAC,Main SL1500, EL_ENC_B	S1 ;S3	2	each	28120014	Block,Mounting,ZB4 BZ009	7
Enclosure, EAC,Main SL1500, EL_ENC_B	S1 ;S3	3	each	28120012	Block,1 x N/O Contact,ZBE101	6;7
Enclosure, EAC,Main SL1500, EL_ENC_B	S2	1	each	28120011	Block,1 x N/C Contact,ZBE102	7
Enclosure, EAC,Main SL1500, EL_ENC_B	S3	1	each	30399462	Keyswitch,2 Position,IP69 Spring Rtn	7

PIUS.SL.SUM_LIST.F02.R09

Assembly	Device tag	Qty	UoM	ERP Number	Description	Page No.
Enclosure, EAC,Main SL1500, EL_ENC_B	TA1	1	each	28263800	Control,V/S/1.0HP/D700/115/1-220/3	5
Enclosure, EAC,Main SL1500, EL_ENC_B	TBA1	1	each	28217685	Power Supply,24VDC/60W/2.5A	5
Enclosure, EAC,Main SL1500, EL_ENC_B	W99	1	each	28218322	Cord,Power/SJWC/9Ft/14/3AWG/15A/125V	5
Enclosure, EAC,Main SL1500, EL_ENC_B	WPE4	1	each	30715597	Strap,Ground,Bond,G/Y,300mm,6.5mm,UL467	5
Enclosure, EAC,Main SL1500, EL_ENC_B	WPE5	1	each	30715741	Strap,Ground,Bond,G/Y,150mm,6.5mm,UL467	5
Enclosure, EAC,Main SL1500, EL_ENC_B	X1	5	each	28326941	Block,Terminal,5.2mm,Feed through	10
Enclosure, EAC,Main SL1500, EL_ENC_B	X1	3	each	30420780	Terminal,End cover,D-ST 2,5	10
Enclosure, EAC,Main SL1500, EL_ENC_B	X2 ...X4	21	each	30378288	Terminal,Push-in,DB Level,0.14 - 4mm	10
Enclosure, EAC,Main SL1500, EL_ENC_B	XU	11	each	30476101	Terminal,End Clamp,CLIPFIX 35-5	10
Enclosure, EAC,Main SL1500, EL_ENC_B	XU	3	each	30389684	Terminal,End cover,PTTB 2.5	10
Enclosure, EAC,Main SL1500, EL_ENC_B	XE1	1	each	28217660	Lug, Grounding/Aluminum	5
Enclosure, EAC,Main SL1500, EL_ENC_B	XE2	1	each	28218378	Kit, Ground Bar/5 Terminal/2.54"L	5
Kit, EAC,Mechanical Misc SL1500	B1	1	each	24106343	Lightbarrier O6P-FPKG/AS/4P	9
Kit, EAC,Mechanical Misc SL1500	B1	1	each	24100158	Reflector,LBR RF7106023	9
Kit, EAC,Mechanical Misc SL1500	B2	1	each	28218159	Sensor,Inductive,PNP,12-48VDC,M12	9
Kit, EAC,Mechanical Misc SL1500	ESD1	1	each	30761990	Enclosure, EAC, E-Stop IP69	6
Kit, EAC,Mechanical Misc SL1500	PJ1	1	each	28279046	Beacon,BASE,24VDC,D60MM,WHITE,5M CABLE	8
Kit, EAC,Mechanical Misc SL1500	PJ1-P2	1	each	28254280	Lamp,Patlite/LR6/Red/Module	8
Kit, EAC,Mechanical Misc SL1500	PJ1-P6	1	each	28254398	Horn,Stacklight/White/Patlite	8
Kit, EAC,Mechanical Misc SL1500	W14	1	each	30517856	Cable,Sensor,IP69k,M8,Straight	9
Kit, EAC,Mechanical Misc SL1500	W15	1	each	28218277	Cable,Photoeye/4-Pin/Euro/90Deg/5mL	9

Parts List



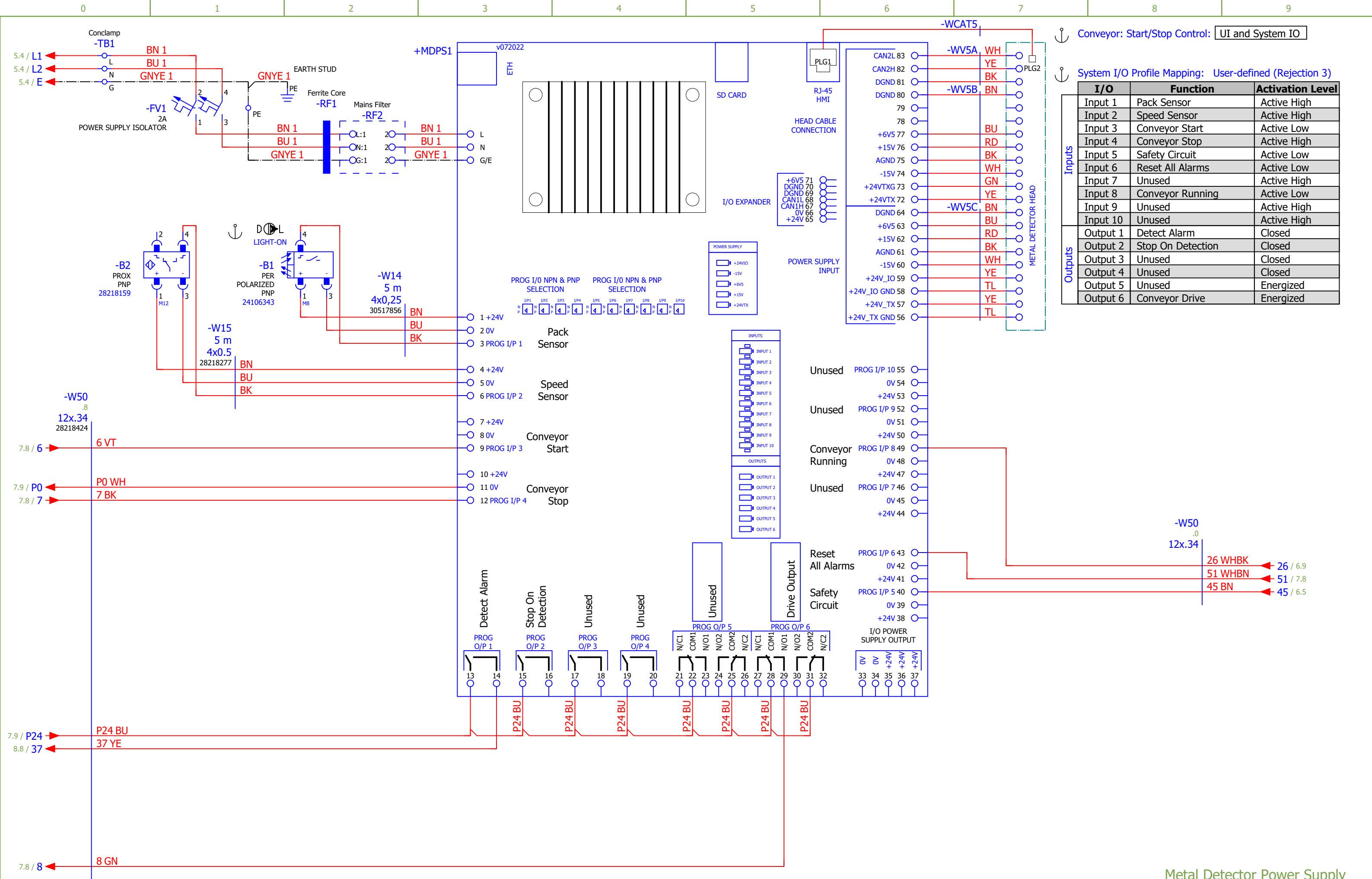


Control Circuits

75055 / CTRL /Start/Stop (SCH) />Pot IP69K (SCH) /Key on Enc /Patlite LR6 IP69 /Red R /D /IntPt(P24,P0,6,7,8,29,37,51)

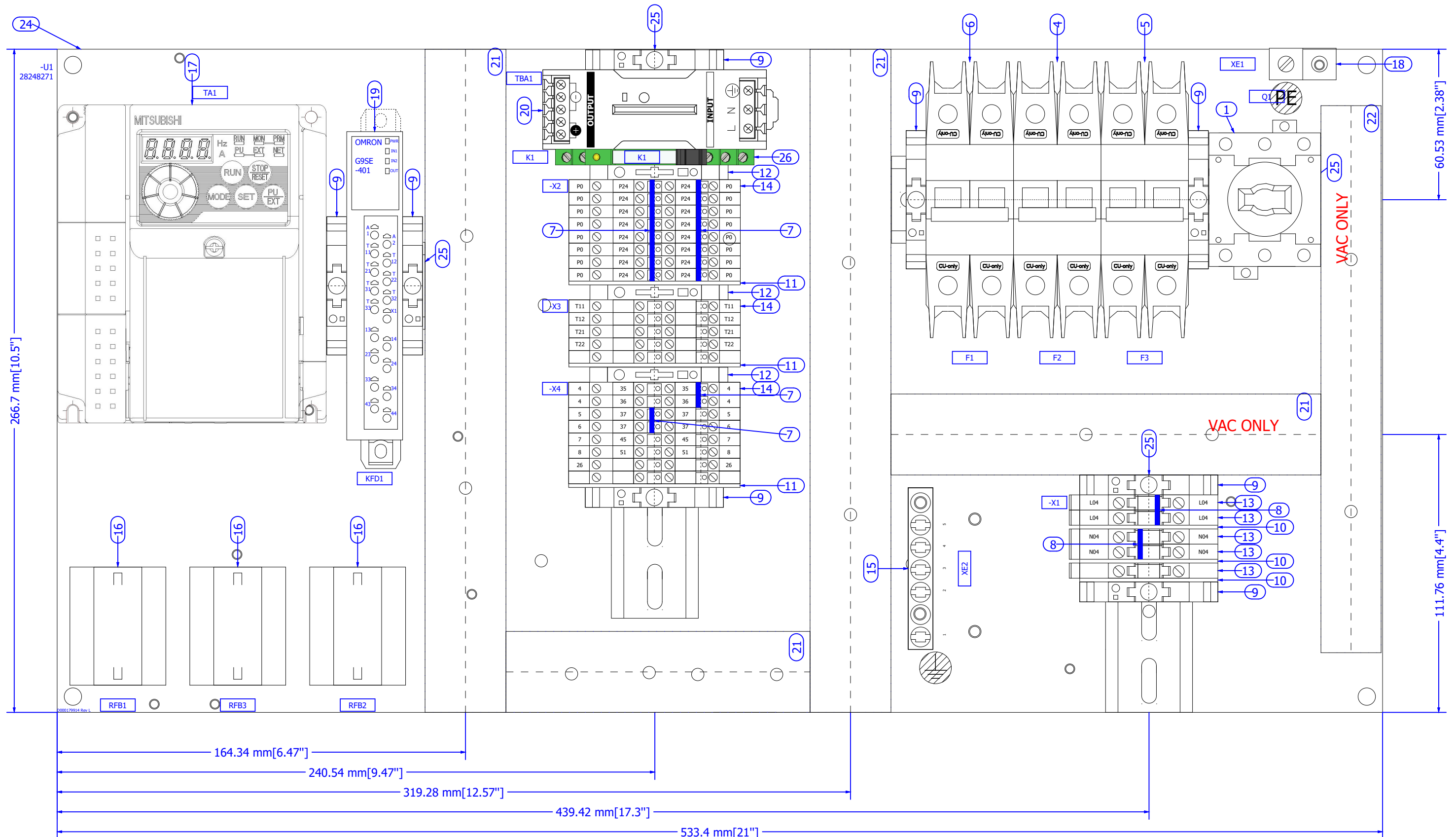
 METTLER TOLEDO	SAFELINE Metal Detection	Mettler-Toledo Product Inspection 1571 Northpointe Parkway Lutz, FL 33558		This drawing and any design, development, invention, or copyright which it may embody or represent are the sole property of Mettler-Toledo. This drawing may not be copied in whole or in part, nor any subject matter shown herein be manufactured, without the prior express written consent from an authorized agent of Mettler-Toledo.			Project Name EPL010605	Project Number EPL010605	Change	9/14/2023	Assembly ID 30916892A		ECN No.	
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							Variant Description Enclosure, EAC,Main SL1500, EL_ENC_B	Checked D. Doyle	9/13/2023	Mounting Location	Project Status	Format ANSI A		
							Page Description Control Circuits	Document Type Electrical Schematic		DCC &EFS1	Project Rev A	Page 7 / 16		





85006 / MDPS /V5 /Standard /PS >PER (IFM) + M8 Cable /SP Prox (SCH) + 5m Cable /I3 > Safety Circuit /> Reset All Alarms /> Detect /> Stop on Detect /IntPt(26,45) /IntPt(P24,P0,6,7,8,29,37,51)

NOTE:
Circuit Breakers shown are A-B 1492-M Series.
Regardless of brand, verify terminal 1 is upwards.



Panel Layout

Cable Connection List

PIUS.SL.CABLE_CON_LIST.R05

Cable DT W99 ☐	Material 28218322	Description Cord,Power/SJWC/9Ft/14/3AWG/15A/125V				Conductors / Cross-section 3x2.5	Cable length A/N
	Cable Location	Label	Color	Cross-section	Target 1 / Page.Column	Target 2 / Page.Column	Remark
			BK	2.5	/5.0	-Q1:L1 /5.0	
			WH	2.5	/5.0	-Q1:L3 /5.0	
Cable DT WCAT5 ☐	Material	Description				Conductors / Cross-section	Cable length A/N
	Cable Location	Label	Color	Cross-section	Target 1 / Page.Column	Target 2 / Page.Column	Remark
					MDPS1 :PLG1 /9.6	:PLG2 /9.7	
Cable DT WV5A ☐	Material	Description				Conductors / Cross-section	Cable length A/N
	Cable Location	Label	Color	Cross-section	Target 1 / Page.Column	Target 2 / Page.Column	Remark
			BK		-MDPS1:81 /9.6	/9.7	
			WH		-MDPS1:83 /9.6	/9.7	
Cable DT WV5B ☐	Material	Description				Conductors / Cross-section	Cable length A/N
	Cable Location	Label	Color	Cross-section	Target 1 / Page.Column	Target 2 / Page.Column	Remark
			BK		-MDPS1:75 /9.6	/9.7	
			BN		-MDPS1:80 /9.6	/9.7	
			BU		-MDPS1:77 /9.6	/9.7	
			GN		-MDPS1:73 /9.6	/9.7	
			RD		-MDPS1:76 /9.6	/9.7	
			WH		-MDPS1:74 /9.6	/9.7	
Cable DT WV5C ☐	Material	Description				Conductors / Cross-section	Cable length A/N
	Cable Location	Label	Color	Cross-section	Target 1 / Page.Column	Target 2 / Page.Column	Remark
			BK		-MDPS1:61 /9.6	/9.7	
			BN		-MDPS1:64 /9.6	/9.7	
			BU		-MDPS1:63 /9.6	/9.7	
			RD		-MDPS1:62 /9.6	/9.7	
			TL		-MDPS1:58 /9.6	/9.7	
			TL		-MDPS1:56 /9.6	/9.7	
			WH		-MDPS1:60 /9.6	/9.7	
			YE		-MDPS1:59 /9.6	/9.7	
			YE		-MDPS1:57 /9.6	/9.7	

☒ = Cable connections terminated ☐ = Add cable connection

Cable Connection List

Cable Connection List

Cable DT WPE5	☐	Material	30715741	Description	Strap,Ground,Bond,G/Y,150mm,6.5mm,UL467				Conductors / Cross-section	Cable length	A/N	
		Cable Location		Label	Color	Cross-section	Target 1 / Page.Column	Target 2 / Page.Column	Remark			
		BEN1				10 mm ²	:PE3 /5.2	:PE5 /5.2				
Cable DT W9	☐	Material	28218425	Description	Cable,AWG/22/3-Cond/Braided Shield				Conductors / Cross-section	3x.34	Cable length	A/N
		Cable Location		Label	Color	Cross-section	Target 1 / Page.Column	Target 2 / Page.Column	Remark			
		BEN1			WH	.50	-R1:B /7.2	-TA1:2 /7.2				
		=			BK	.50	-R1:A /7.2	-TA1:5 /7.3				
		=			RD	.50	-R1:C /7.2	-TA1:10 /7.2				
		=			SH	.34	BEN1 -W9 /7.2	-TA1:5 /7.3				
Cable DT W10	☐	Material	28218424	Description	Cable,22AWG/12 Conductors/Sheilded				Conductors / Cross-section	12x.34	Cable length	A/N
		Cable Location		Label	Color	Cross-section	Target 1 / Page.Column	Target 2 / Page.Column	Remark			
		BEN1		7	BK	0.34	-X4:7 /7.4	-S2:12 /7.4				
		=			BN	0.34						
		=		6	RD	0.34	-X4:6 /7.4	-S1:14 /7.4				
		=		5	OG	0.34	-S1:13 /7.4	-X4:5 /7.4	Double Ferrule			
		=			YE	0.34						
		=			GN	0.34						
		=			BU	0.34						
		=			VT	0.34						
		=			GY	0.34						
		=			WH	0.34						
		=		P24	WHBK	0.34	-X2:P24 /7.5	-S3:13 /7.5				
		=		51	WHBN	0.34	-X4:51 /7.5	-S3:14 /7.5				
		=			SH	0.34						
Cable DT W12	☐	Material	28218244	Description	Cable,AWG/22/6-Cond/TC/Pharm				Conductors / Cross-section	6x0.34	Cable length	A/N
		Cable Location		Label	Color	Cross-section	Target 1 / Page.Column	Target 2 / Page.Column	Remark			
		BEN1		T12	BK	.34	-SGC1:12 /6.0	-X3:T12 /6.4				
		=			WH	.34						
		=		T22	RD	.34	-SGC1:22 /6.0	-X3:T22 /6.4				
		=		T21	GN	.34	-SGC1:21 /6.0	-X3:T21 /6.3				
		=		T11	OG	.34	-SGC1:11 /6.0	-X3:T11 /6.3				
=			BU	.34								

☒ = Cable connections terminated

☐ = Add cable connection

Cable DT W41	☐	Material	28218426	Description	Cable,AWG/20/2-Cond/CC600			Conductors / Cross-section	2x.50	Cable length	A/N
		Cable Location		Label	Color	Cross-section	Target 1 / Page.Column	Target 2 / Page.Column		Remark	
		BEN1		T31	BK(1)	.50	-KFD1:T31 /6.4	-S3:23 /6.5			
		=		T32	BK(2)	.50	-KFD1:T32 /6.4	-S3:24 /6.5			
Cable DT W14	☐	Material	30517856	Description	Cable,Sensor,IP69k,M8,Straight			Conductors / Cross-section	4x0,25	Cable length	5 m
		Cable Location		Label	Color	Cross-section	Target 1 / Page.Column	Target 2 / Page.Column		Remark	
		MDPS1			BN	0,25	-B1:1 /9.2	-MDPS1:1 /9.3			
		=			WH	0,25					
		=			BU	0,25	-B1:3 /9.2	-MDPS1:2 /9.3			
		=			BK	0,25	-B1:4 /9.2	-MDPS1:3 /9.3			
Cable DT W15	☐	Material	28218277	Description	Cable,Photoeye/4-Pin/Euro/90Deg/5mL			Conductors / Cross-section	4x0.5	Cable length	5 m
		Cable Location		Label	Color	Cross-section	Target 1 / Page.Column	Target 2 / Page.Column		Remark	
		MDPS1			BN	.5	-B2:1 /9.1	-MDPS1:4 /9.3			
		=			WH	.5					
		=			BU	.5	-B2:3 /9.1	-MDPS1:5 /9.3			
		=			BK	.5	-B2:4 /9.1	-MDPS1:6 /9.3			

☒ = Cable connections terminated

☐ = Add cable connection

Wire Connection List

PIUS.SL.CON_LIST.F27.R14

Label	Color	Cross-section	Target 1	Target 2	Remark	/Page .Column
L01	☒	BK	2.5	Q1:T1	RFA1:P	/5.0
L02	☒	BK	2.5	RFA1:P'	F1:1	/5.0
L03	☒	BK	2.5	F1:2	F2:1 Double Ferrule	/5.0
L03	☒	BK	2.5	F2:1	F3:1 Double Ferrule	/5.2
N01	☒	WH	2.5	Q1:T3	RFA1:N	/5.0
N02	☒	WH	2.5	RFA1:N'	F1:3	/5.0
N03	☒	WH	2.5	F2:3	F1:4 Double Ferrule	/5.2
N03	☒	WH	2.5	F3:3	F2:3 Double Ferrule	/5.6
	☒	GNYE	2.5	XE2:1	RFA1:PE	/5.1
P24	☒	BU	1	X2:P24	TBA1:V+	/5.2
L04	☒	BK	2.5	X1:L04	TBA1:L	/5.2
L04	☒	BK	2.5	F2:2	X1:L04	/5.2
P0	☒	WHBU	1	TBA1:V-	X2:P0	/5.2
N04	☒	WH	2.5	TBA1:N	X1:N04	/5.2
N04	☒	WH	2.5	X1:N04	F2:4	/5.2
	☒	GNYE	2.5	TBA1:PE	XE2:2	/5.3
L05	☒	BK	2.5	F3:2	TA1:R/L1	/5.6
N05	☒	WH	2.5	F3:4	TA1:S/L2	/5.6
	☒	GNYE	2.5	TA1:PE	XE2:4	/5.7
P0	☒	WHBU	1	TA1:SD	X2:P0 x3 Loops @RFB3	/6.0
23	☒	BU	1	KFD1:34	TA1:S1 x3 Loops @RFB3	/6.0
22	☒	BU	1	KFD1:33	KFD1:43 Double Ferrule	/6.0
22	☒	BU	1	KFD1:43	TA1:SC x3 Loops @RFB3	/6.1
25	☒	BU	1	KFD1:44	TA1:S2 x3 Loops @RFB3	/6.1
T21	☒	BU	1	X3:T21	KFD1:T21	/6.3
T11	☒	BU	1	X3:T11	KFD1:T11	/6.3
P0	☒	WHBU	1	KFD1:A2	X2:P0	/6.3
P24	☒	BU	1	X2:P24	KFD1:A1	/6.3
T22	☒	BU	1	KFD1:T22	X3:T22	/6.4

T12	☒	BU	1	X3:T12	KFD1:T12		/6.4
P24	☒	BU	1	X2:P24	KFD1:T33		/6.4
45	☒	BU	1	KFD1:X1	X4:45		/6.4
4	☒	BU	1	KFD1:14	X4:4		/6.6
P24	☒	BU	1	X2:P24	KFD1:13		/6.6
26	☒	BU	1	TA1:A	X4:26		/6.8
4	☒	BU	1	X4:4	TA1:C		/6.8
P24	☒	BU	1	X2:P24	TA1:SE		/7.3
8	☒	BU	1	X4:8	TA1:STF	x3 Loops @RFB3	/7.4
5	☒	OG	.34	S2:11	S1:13	Double Ferrule	/7.4
P24	☒	BU	1	X2:P24	X4:5		/7.4
P24	☒	BU	1	X2:P24	X2:P24		/7.7
P24	☒	BU	.34	MDPS1:13	MDPS1:15		/9.3
P24	☒	BU	.34	MDPS1:15	MDPS1:17		/9.3
P24	☒	BU	.34	MDPS1:17	MDPS1:19		/9.4
P24	☒	BU	.34	MDPS1:19	MDPS1:22		/9.4
P24	☒	BU	.34	MDPS1:22	MDPS1:25		/9.5
P24	☒	BU	.34	MDPS1:25	MDPS1:28		/9.5
P24	☒	BU	.34	MDPS1:28	MDPS1:31		/9.5
35	☒	BU	1	K1:14	X4:35		/8.1
36	☒	BU	1	K1:11	X4:36		/8.1
P0	☒	WHBU	1	K1:A2	X2:P0		/8.7
37	☒	BU	1	X4:37	K1:A1		/8.7

☒ = Wire connection terminated ☐ = Add wire connection

Wire Connection List



SAFELINE
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This safety circuit diagram is an outcome of a comprehensive risk assessment and qualification of the circuit components and architecture. This safety circuit should not be modified in any manner at any time without the express written consent of Mettler-Toledo. Modification of the circuit structure or components may cause damage to property, injury to persons, or death.

Project Name EPL010605	Project Number EPL010605	Change	9/26/2023	Assembly ID 30916892A	ECN No.
Project Description M34R V5 Metal Detector SL1500	Drawn E. Ymeraj	9/13/2023	Function	Scale 1 : 1	
Variant Description Enclosure, EAC,Main SL1500, EL_ENC_B	Checked D. Doyle	9/13/2023	Mounting Location	Project Status	Format ANSI A
Page Description Wire Connection List	Document Type Electrical Schematic	DCC &EFS1	Project Rev A	Page 16 / 16	